Revenue & User Analysis Report

1) Introduction

This project analyzes the **revenue performance and user behavior dynamics** of a gaming platform using transactional payment data. The primary goal is to evaluate not only the growth of revenue, but also the underlying drivers of that growth, including user acquisition, retention and monetization efficiency.

The analysis focuses on answering the following key business questions:

- What factors drive changes in Monthly Recurring Revenue (MRR)?
- Is the observed growth sustainable over time?
- How do user acquisition and churn dynamics impact overall performance?
- What is the long-term value of users (LTV), and how does it evolve?

Understanding these aspects is critical for identifying risks in the business model and proposing strategies for sustainable growth.

2) Data & Methodology

2.1 Data Sources

The analysis is based on two main tables:

1) **Project.games_payments**
    - user_id
    - game_name
    - payment_date
    - revenue_amount_usd
2) **Project.games_paid_users**
    - user_id
    - language
    - age

These tables were joined to combine transactional data with user attributes.

2.2 Data Preparation

SQL was used to transform raw data into a structured analytical dataset:

- Monthly revenue per user was calculated using aggregation
- Time-based user behavior was analyzed using window functions
    - LAG() → previous month revenue
    - LEAD() → next month revenue

2.3 User Classification

Users were categorized into lifecycle segments based on revenue changes:

- **New Users** → users making their first payment
- **Expansion Users** → users increasing their spending
- **Contraction Users** → users decreasing their spending
- **Churned Users** → users who stop paying
- **Back From Users** → users returning after inactivity

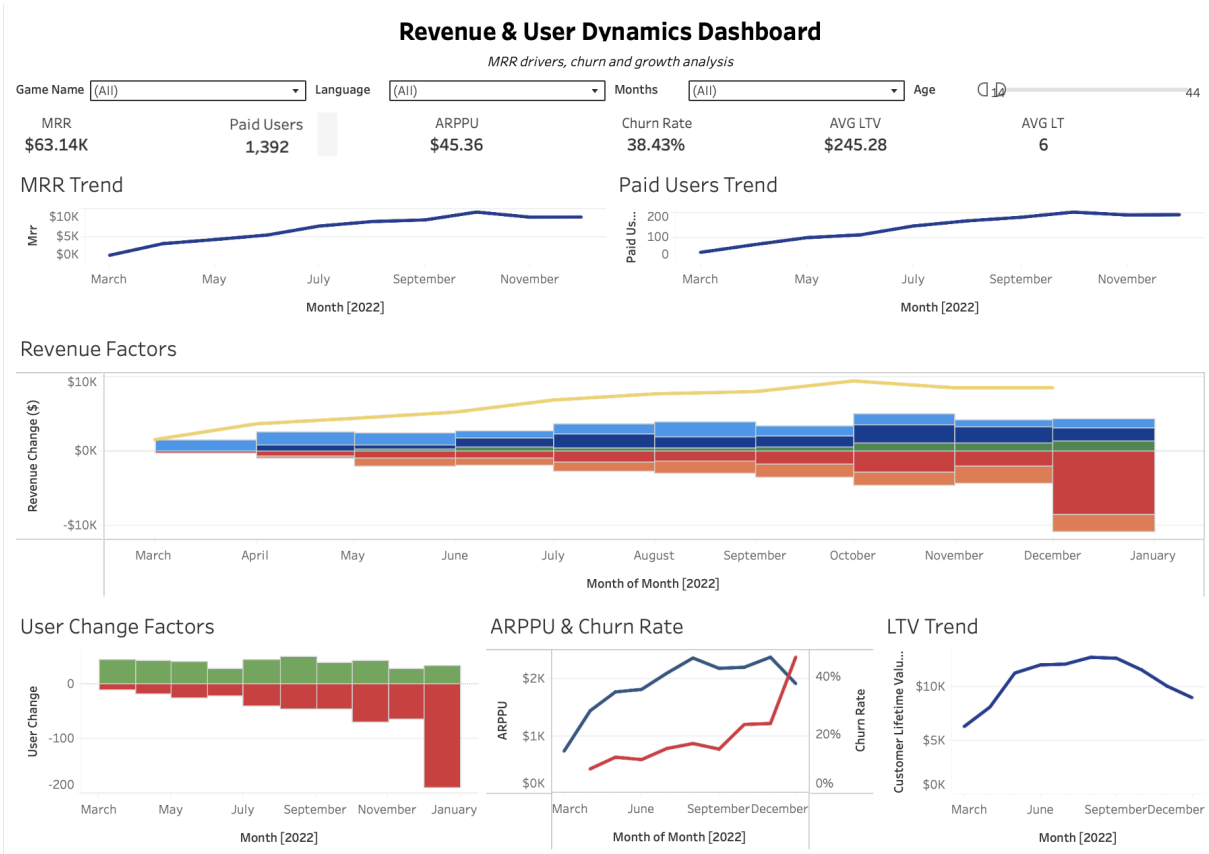
These classification enables detailed analysis of revenue drivers.

2.4 Metrics

- MRR (Monthly Recurring Revenue)
- Paid Users
- ARPPU (Average Revenue Per Paying User)
- Churn Rate
- Revenue Churn Rate
- LTV (Lifetime Revenue)
- LT (Customer Lifetime)

3) Dashboard Overview

The analysis was visualized using a Tableau dashboard that includes KPIs multiple analytical charts.



The dashboard includes:

- KPI summary (MRR, Paid Users, ARPPU, Churn Rate, LTV, LT)
- Revenue decomposition (Revenue Factors)
- User dynamics (User Change Factors)
- Monetization analysis (ARRPU & Churn Rate)
- Long-term value trends (LTV Trend)

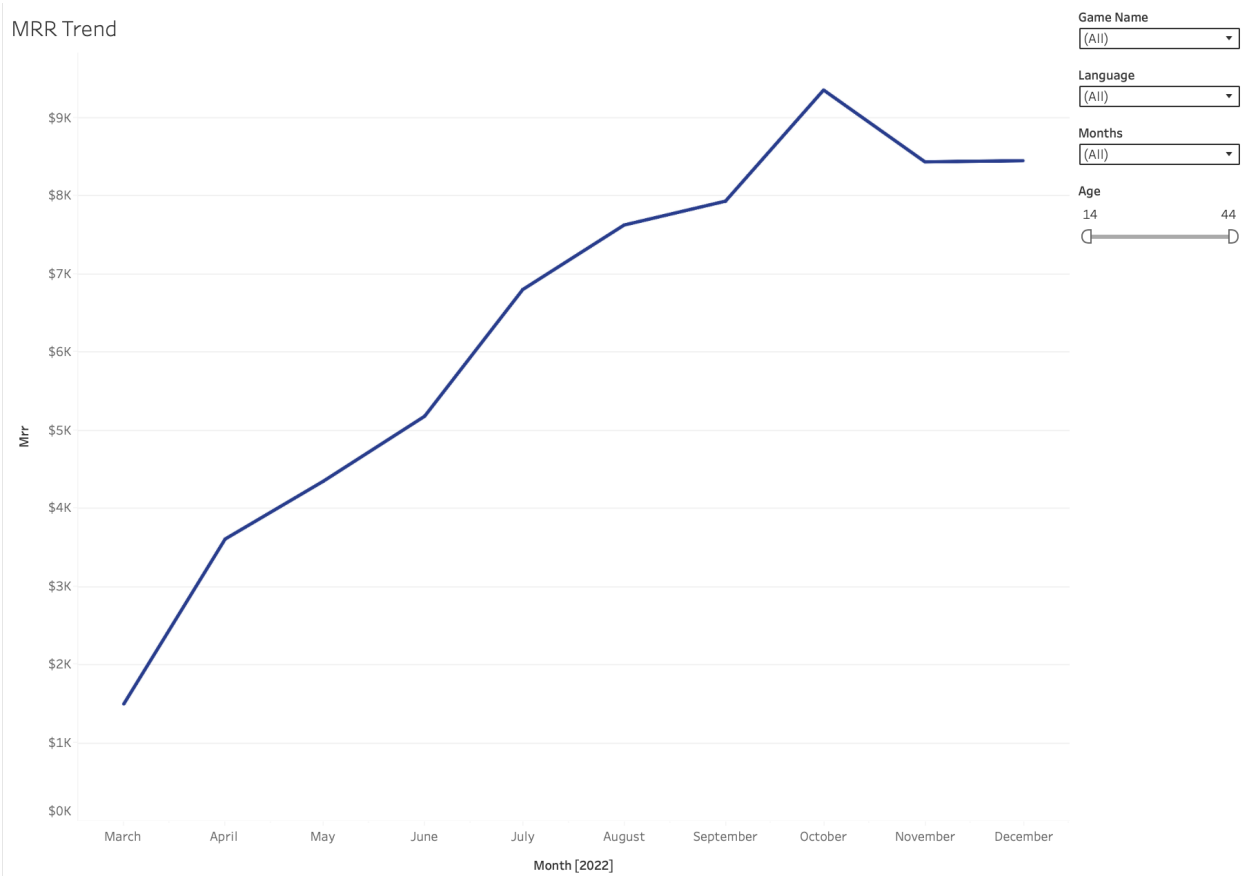
4) Analysis

4.1 MRR Trend

MRR demonstrates a strong upward trend from the beginning of the observed period until approximately October, indicating successful revenue growth.

This increase suggests that the platform is effectively acquiring users and/or increasing spending per user. However, after reaching its peak, MRR begins to stabilize and slightly decline towards the end of the period.

This change in trend indicates that the growth is not fully sustainable and may be impacted by rising churn or reduced user retention.

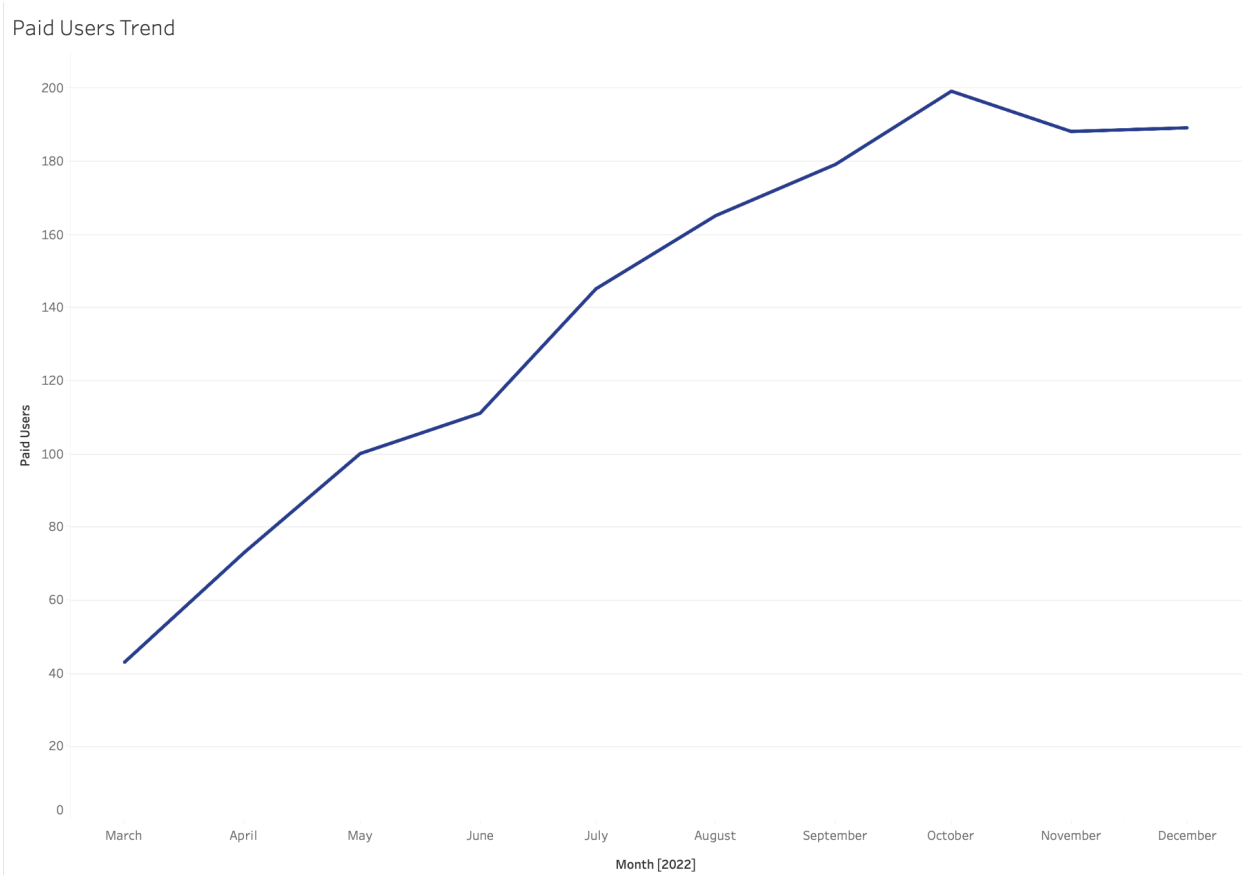


4.3 Paid Users Trend

The number of paid users increases steadily during the initial growth phase, reflecting successful user acquisition strategies.

However, in later months, the growth in paid users slows down and becomes more volatile. This pattern aligns with the increase in churn observed in other metrics.

This suggests that although the platform continues to attract new users, it struggles to retain them over time.



4.3 Revenue Factors

Revenue decomposition provides deeper insight into the sources of revenue change.

Positive contributions are mainly driven by:

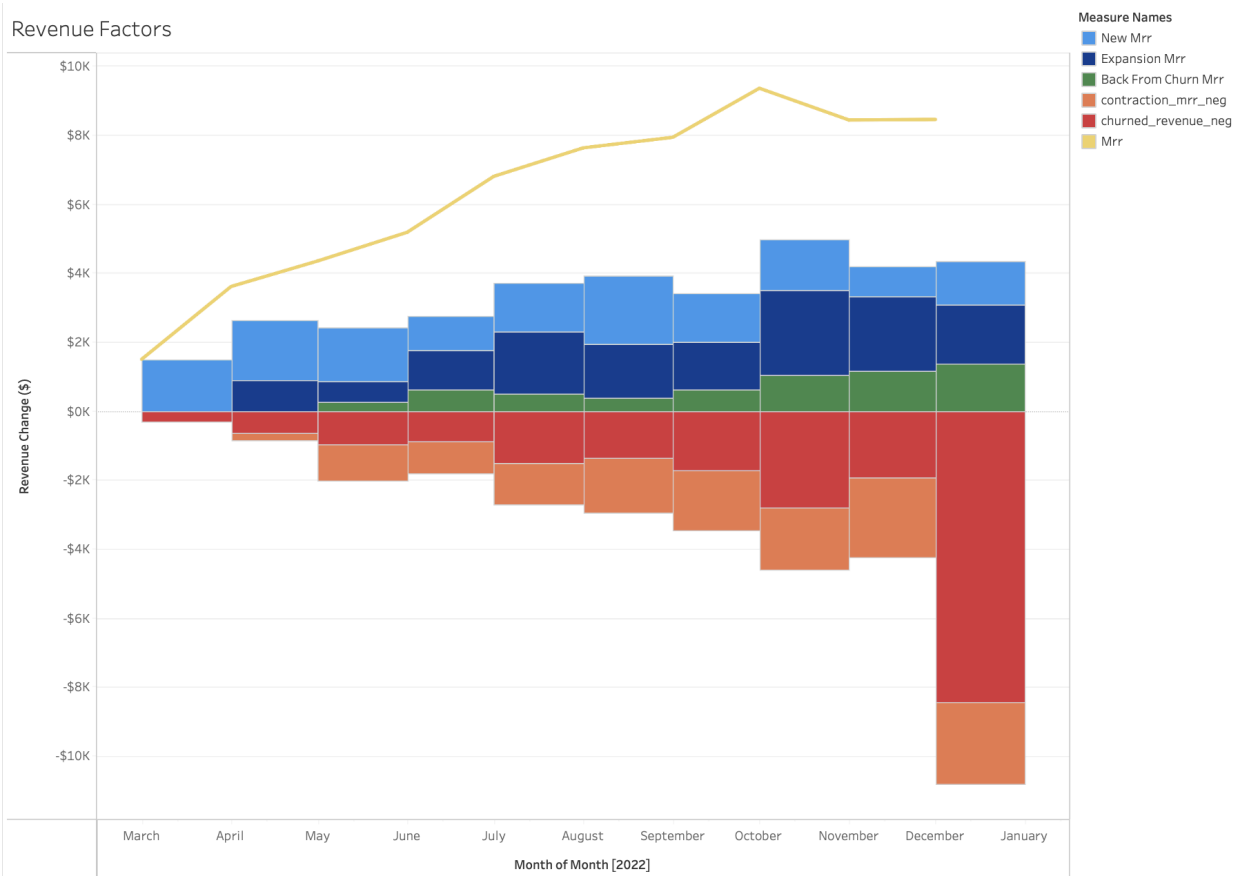
- New users entering the platform
- Existing users increasing their spending (expansion)

However, negative components become increasingly significant:

- Churned revenue grows sharply
- Contraction also contributes negatively.

In the later months, the negative effects of churn and contraction begin to offset the gains from new and expanding users.

This indicates that revenue losses are accelerating and represent a major risk to sustained growth



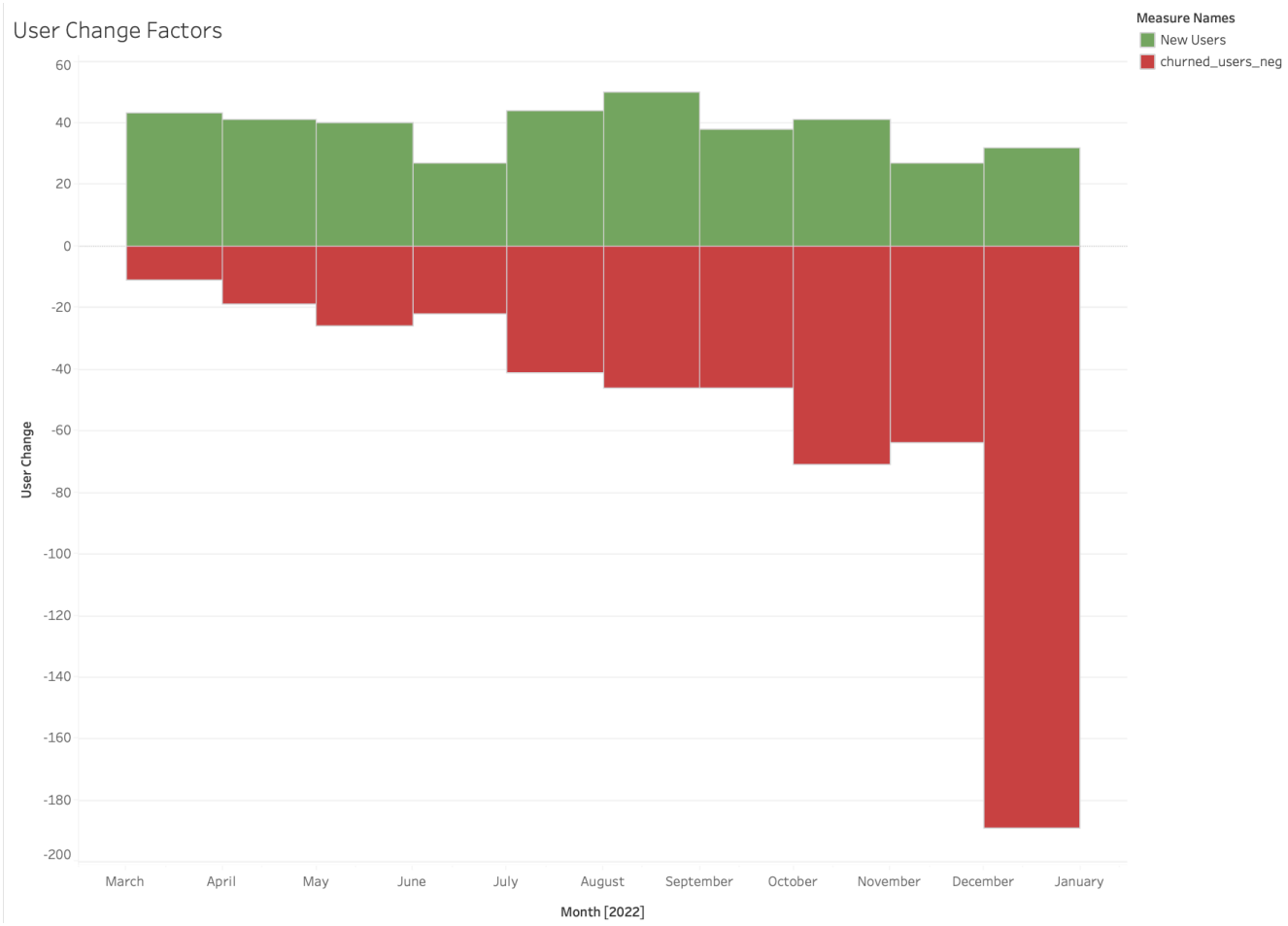
4.4 User Change Factors

The analysis of user dynamics reveals that:

- New user acquisition remains relatively stable over time
- Churned users increase significantly

By the end of the period, churn surpasses new user acquisition, leading to negative net user growth.

This indicates a structural issue in user retention, where the platform is unable to maintain its user base despite continuous acquisition efforts.



4.5 ARPPU & Churn Rate

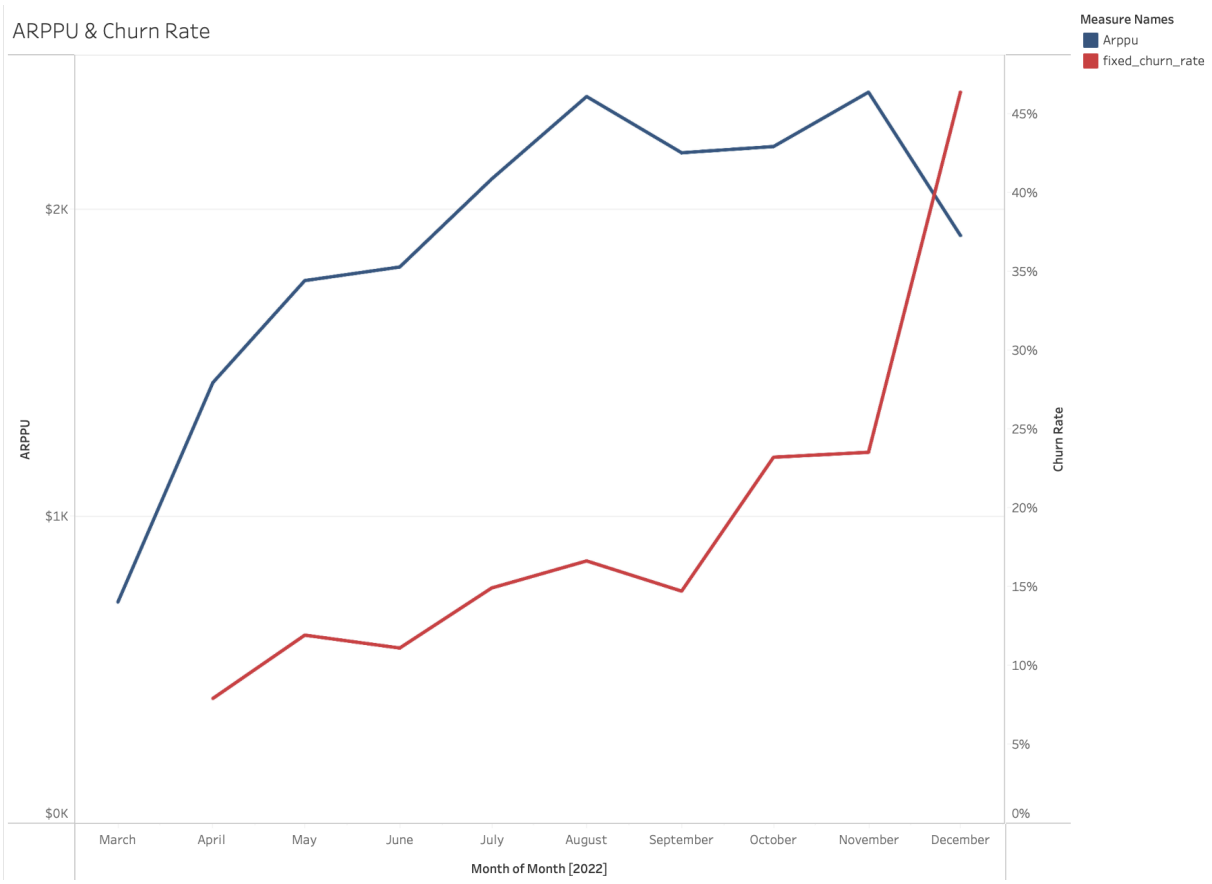
ARPPU shows a consistent upward trend, indicating improved monetization efficiency. Users who remain on the platform are spending more over time.

However, churn rate also increases significantly, reaching high levels in later months.

This creates a critical imbalance:

- Higher revenue per user
- Lower user retention

This suggests that monetization improvements are not sufficient to compensate for increasing user loss.

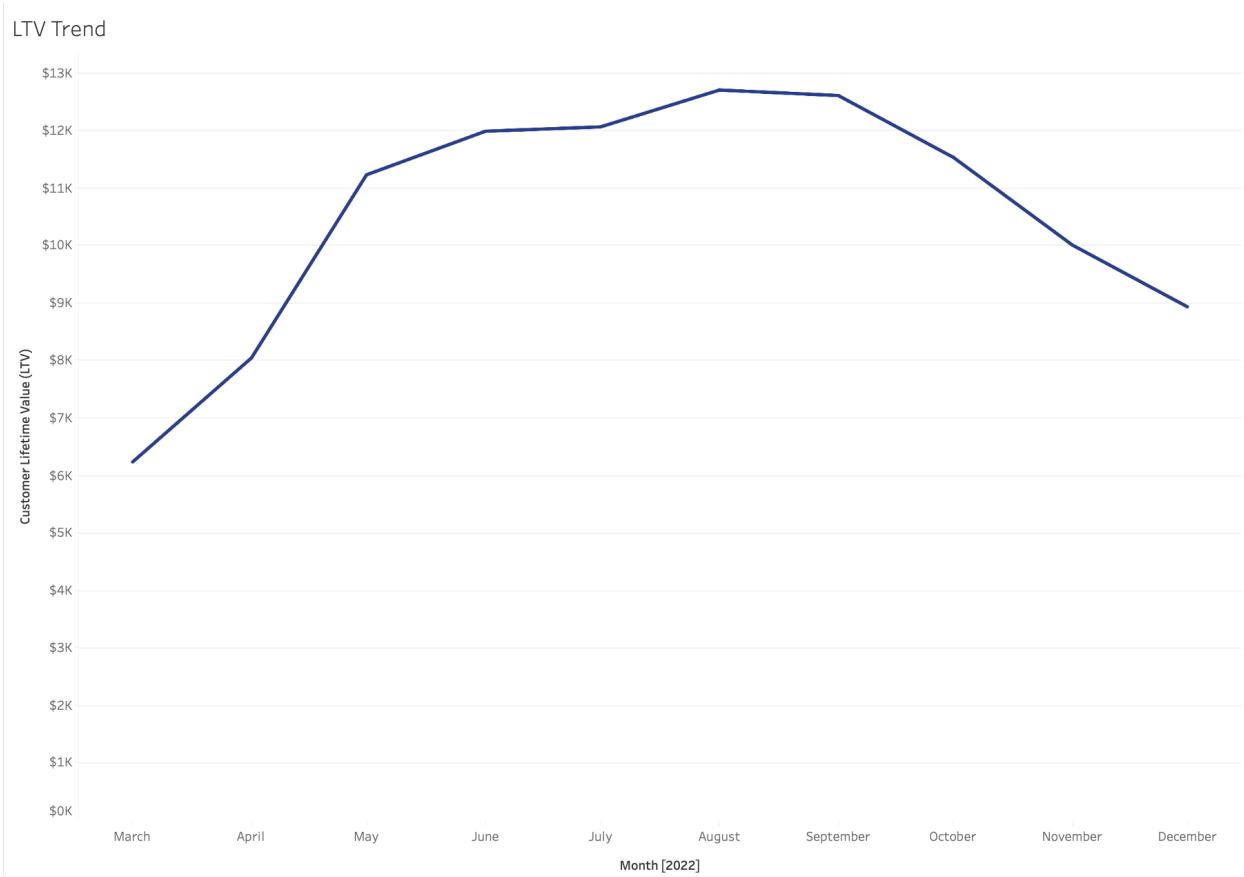


4.6 LTV Trend

LTV increases during the early stages of the period, reflecting strong revenue generation and relatively stable retention.

However, as churn increases, LTV begins to decline. This indicates that users are not staying long enough to generate maximum value.

The decline in LTV highlights the negative impact of poor retention on long-term revenue potential.



4.6 Customer Lifetime (LT)

The average customer lifetime is approximately 6 months, indicating that users engage with the platform for a moderate duration.

However, this duration is not sufficient to maximize long-term value.

Increasing customer lifetime would directly improve LTV and overall business performance.

5) Key Insights

- Revenue growth is initially strong but slows down over time
- Growth is driven by new users and expansion
- Churn increases significantly and becomes the dominant negative factor
- Paid user growth becomes unstable
- ARPPU increases, but retention worsens
- LTV declines due to reduced customer lifetime

6) Recommendations

6.1 Improve Retention (Highest Priority)

- Identify churned user segments
- Improve onboarding and engagement
- Implement retention strategies

6.2 Re-engage Churned Users

- Develop targeted win-back campaigns
- Offer personalized incentives

6.3 Leverage High-Value Users

- Focus on users with high ARPPU
- Introduce premium features and upselling strategies

6.4 Balance Growth Strategy

Shift from an acquisition-heavy approach to a balanced strategy that emphasizes both retention and monetization

7) Conclusion

The platform demonstrates strong initial growth but faces significant challenges in user retention.

While monetization per user improves, increasing churn reduces overall sustainability.

Long-term success depends on:

- Reducing churn
- Increasing customer lifetime
- Stabilizing revenue drivers

Without these improvements, growth will likely plateau or decline.